

Question				Marking details	Marks available																	
					AO1	AO2	AO3	Total	Maths	Prac												
5	(a)			<table><tr><td></td><td>Properties are fixed</td><td>Properties can vary</td></tr><tr><td>copper</td><td>✓</td><td></td></tr><tr><td>brass</td><td></td><td>✓</td></tr><tr><td>bronze</td><td></td><td>✓</td></tr></table> award (2) for all three correct award (1) for any two correct		Properties are fixed	Properties can vary	copper	✓		brass		✓	bronze		✓			2	2		
	Properties are fixed	Properties can vary																				
copper	✓																					
brass		✓																				
bronze		✓																				
	(b)			copper, bronze and brass are all metal alloys ☐ copper and bronze are both metal alloys ☐ copper and brass are both metal alloys ☐ bronze and brass are both metal alloys ☒			1	1														
	(c)			B			1	1														

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			no – must give reason brass and bronze are used in electrical industry / electrical contacts / have uses that depend on conducting electricity brass and bronze contain metals and therefore conduct electricity brass and bronze have delocalised electrons (and therefore conduct electricity) accept reference to only one of brass / bronze neutral answers brass and bronze are metals brass and bronze conduct electricity			1	1		
				Question 5 total	0	0	5	5	0	0